HumanBench: Can AI Write Like a Human?

Pragnyan Ramtha
pragnyanramtha@agent7.dev

Abstract

HumanBench v0.1 is one benchmark of everyday writing with one leaderboard. An answer counts only if it passes three gates in order: task checks run by scripts, a quality review in which a language model judges a minimum quality bar, and AI detection by a named open classifier at a cutoff calibrated on 801 human texts. Each card's score is the share of its three answers passing all gates; the model's score is the mean over all cards, with a bootstrap interval over cards. Public and private scores are secondary columns. There are 212 cards: 132 public (52 built from real writing and 80 invented scenarios) and 80 private. Across eleven models, overall scores range from 1.9% to 23.1%, with wide intervals. Answer quality over all cards, the share passing the first two gates, ranges from 27.0% to 83.3%. HumanBench measures task compliance, a model-judged quality bar and detectability under named open classifiers; no human readers were involved. Its results depend on the classifier, the card mix, each model's reasoning effort and how it was accessed, and models under test helped draft, audit and judge the benchmark. We release public cards, code and a text-free scoring ledger at https://github.com/pragnyanramtha/human-bench.

1 Introduction

People ask language models to write text that they will send under their own name: an email to a supplier, a reply in a forum thread, a product review, a toast, a postmortem. Two properties matter for this use. The text has to do the job that was asked. The text should also not be marked as machine-written by the classifiers that are now run on such text. HumanBench v0.1 measures both properties on the same answers.

Existing evaluations cover the two properties separately. Detector benchmarks hold a set of texts fixed and score detectors on it [6, 10, 15, 28]. They show how well a detector works. They do not show whether a given model can complete a writing task without being flagged, and a text can lower a detector's score by getting worse: the attacks studied in RAID include misspellings and character substitutions [6]. Writing benchmarks score quality, preference or surface style with rubrics, model judges or pattern counts [8, 24, 30]. They do not ask whether a classifier flags the text, and an answer can satisfy a rubric while being written in a model's default style. Detectability alone can be raised by degraded text. A rubric alone mostly reports how well a model follows instructions.

HumanBench joins the two with a conjunction. Each task is a *card*: a request as a user would type it, the user's notes, and hidden lists of what the answer must contain, convey and keep out. A model answers each card three times. Each answer passes through three gates in order (Figure 1): task checks run by scripts, a quality review by a language model acting as judge, and AI detection by an open classifier at a cutoff that 99% of a pool of human texts stay under. The HumanBench score is the share of answers that pass all three. There are two kinds of card. A *reference-matched* card was built backwards from a real piece of writing by a real person, and that piece passes the task checks, the quality review by the main judge and AI detection. An *invented scenario* was drafted by a model, and this kind also covers thread replies and simulated workspaces. The benchmark has 212 cards. The 132 public cards are 52 reference-matched cards and 80 invented scenarios. The 80 private cards are invented scenarios that are held back. Every model gets one score over all cards. Beside it, the public score lets anyone reproduce that part; the private score shows whether it holds on cards no model can have been trained or tuned on. The parts are not expected to be equal: the public part also contains the harder reference-matched cards. The overall score depends on the mix of 52 reference-matched cards and 160 invented scenarios.

What the benchmark does not measure. No human readers were involved at any stage. In this paper, "reads as human" means only that a named open classifier did not flag the text at a calibrated cutoff. Studies with human judges show that the verdict depends on who judges [5, 12, 25], so a classifier result is not a statement about readers. The benchmark measures task compliance, a minimum quality bar judged by a model, and detectability under named classifiers. Every claim below stays inside that.

Contributions. (1) A protocol with three gates, one score over all cards, secondary public and private scores, and a companion quality number, with the reasons for not adding the parts as weighted points (Section 3). (2) Reference-matched cards built backwards from archival human text, with a leakage screen and a memorisation probe (Section 4). (3) Measurements of the scoring itself: judge validation, the disagreement between the two judges on identical outputs, classifier calibration, a device check and a human baseline (Section 5). (4) Re-

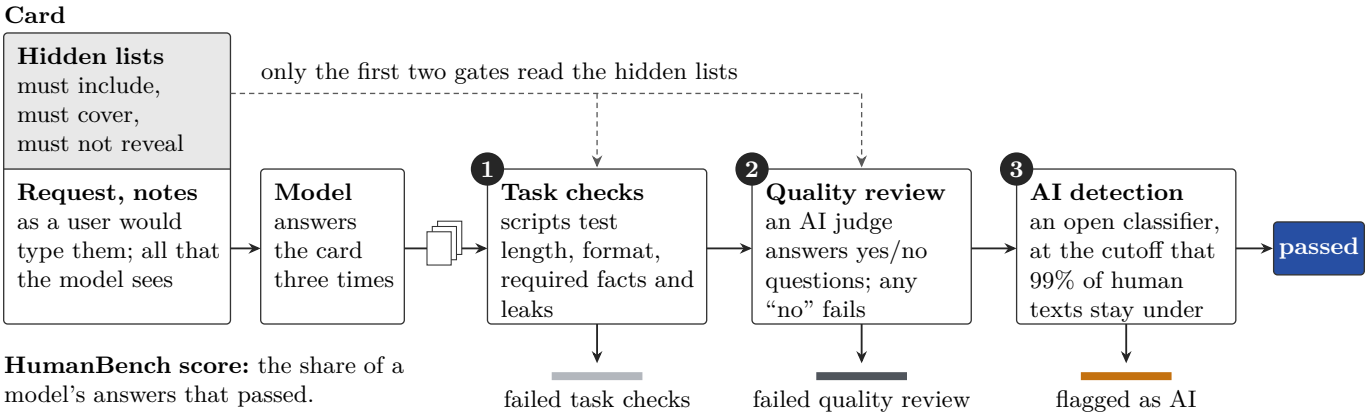


Figure 1. The protocol. A card holds a request and notes, which the model sees, and hidden lists, which only the first two gates read. The model answers each card three times, and each answer goes through the gates in order and stops at the first it fails. Look at the four endings: every answer has exactly one outcome, and only an answer that passes all three gates counts toward the score.

sults for eleven models, broken down by outcome, by classifier and by kind of failure, with a ledger from which the scores can be recomputed without any task or model text (Sections 6 to 8).

2 Related Work

Detector benchmarks hold texts fixed and compare classifiers across generators and domains [6, 10, 15, 28]. HumanBench instead holds classifiers fixed and evaluates task-completing generation. Detector results depend on domain, instruction tuning, language background and paraphrasing [1, 17, 18, 31]. EditLens estimates the extent of model editing rather than binary authorship [26]; its open checkpoints are distinct from Pangram's commercial detector [7, 9]. Zero-shot alternatives use token statistics [2, 11].

WritingBench uses task-specific criteria, LitBench studies model agreement with human story preferences, and SlopBench measures surface behaviours [8, 24, 30]. Distributional studies find differences in grammar, vocabulary and register [14, 20–22, 29]. HumanBench gates detectability on task compliance and minimum quality; only one kind of card is tied to archival human writing.

Human assessments depend on the readers and task [4, 5, 12, 25]; classifier verdicts cannot stand in for them. Benchmark-practice studies motivate uncertainty estimates and reproducibility [3, 16, 23]. We resample over cards rather than treating repeated answers as independent tasks [19]. Evidence of model use by crowd workers motivates archival sourcing [27].

3 Benchmark Design

3.1 Cards, kinds and parts

A card asks for one piece of everyday writing of 250 to 4,000 words, or for several messages that total that much. There are two kinds of card, and every card is scored the same way.

An *invented scenario* is a situation and persona drafted by a model. This kind has eight channels: email, chat, forum, social post, review, blog, work document and personal writing. It has four modes. In *direct* mode the model gets a request and notes. In *thread reply* mode it also gets the thread it answers. In *workspace* mode it gets a simulated inbox, forum or chat as text and must decide what to write and where to send it. *Ghostwriting* is workspace mode with samples of the user's past writing. Each invented scenario has a model-written reference solution, which is used only to prove that the card's checks can be met and carries no human claim.

A *reference-matched* card is built backwards from a real piece of writing (Section 4). The model never sees that piece. These cards are all in direct mode. The 52 used here cover six channels: 16 review, 10 email, 9 personal, 8 forum, 5 blog and 4 work document. There are no chat or social cards of this kind, because no suitable public source of long private messages or posts was identified.

The benchmark has two parts. The public part has 132 cards: the 52 reference-matched cards and 80 invented scenarios, ten per channel (36 direct, 13 thread reply, 23 workspace and 8 ghostwriting). Anyone can read these cards and check a score on them. The private part has 80 invented scenarios of the same kinds that are not released, so that no model can be trained or tuned on them. The two parts are not expected to give equal scores, because only the public part contains the reference-matched cards, which are harder.

3.2 How a card is built and audited

A card holds a prompt in the user's voice, a notes pack (facts, anecdotes, opinions and wants), a few facts about the persona, a length band, and hidden lists that the model never sees. `must_include` holds a few literal facts with accepted spellings. `must_cover` holds plain statements of what the piece has to get across, judged by meaning. `must_not_reveal` holds what the user asked to keep out. Workspace cards add routing checks: required

messages and their targets, barred targets, a limit on the number of messages, and items marked private. Cards contain the ordinary wishes a user would state, such as "keep it light". They contain no instruction to sound human and no advice on evading detection. Figure 2 shows a public invented scenario, and Appendix C a reference-matched card.

Cards are drafted by a model: an invented scenario from a slot in a task matrix, a reference-matched card from a human text. Each card must pass mechanical tests: its reference scores 1.0 on the card's own task checks, and empty outputs, assistant-style wrappers and echoes of the prompt score 0.0. A second model then audits the card against a fixed rubric that covers the fairness of the checks, ambiguity, consistency, realism, the material given, the reference, safety, and evidence from trial runs. The card is fixed and audited again until no finding is rated blocker or major. A card that cannot be made fair is dropped and the drop is recorded. A verdict is bound to a fingerprint of the card's content, so an edited card has no valid audit until it is audited again. The repository records these model audits. It records no card-by-card review by a person (Section 8).

The pre-release review edited 27 public invented scenarios and 20 cards built from real writing, correcting a possible real phone number, names shared with well-known people, street addresses, date and arithmetic slips, visibly appended requests and unsupported persona details. Seven cards built from real writing were dropped for sensitive details about identifiable people, unsuitable content, or inability to make the task fair without those details. Every edited card was audited again until clean, and all eleven models answered it again three times.

3.3 The fixed instruction

Every model receives the same system instruction, reproduced in Appendix A. It says that the person will send or post the text as their own words, asks the model to write so that it sounds natural and human, and asks it to still do everything requested. This is the only place where a model is told the goal. An earlier design used a bare assistant prompt as its headline condition. It was replaced because a model that is not told the goal is tested on its defaults and not on what it can do.

3.4 The three gates

Figure 1 shows the order of the gates. An answer stops at the first gate it fails, so it ends in exactly one of four outcomes: failed task checks, failed quality review, flagged as AI, or passed.

Gate 1: task checks. Scripts, with no model involved, check the answer against the card. The length must fall in the band the request states; when the request says "roughly", a tolerance of 10% applies at both ends. The format must fit the channel: headings and tables fail in email, chat, forum, social, review and personal writing, bold fails in social, review and personal writing, and bullet lists fail in personal writing, unless the card overrides the rule. Assistant talk and placeholders fail, such as an opening that introduces a draft, a bracketed name slot or a closing offer to revise. Required facts must appear and barred phrases must not, after numbers, dates and times are normalised. Hidden characters and text addressed to a grader fail. An answer may not share a run of 13 words with a reference text, and at most 25% of it may be copied from the task's own material. Workspace answers are also checked for routing, leaks of private items and invented money amounts. An answer passes only if every check passes.

Gate 2: quality review. Task checks cannot tell whether a piece does the task, since any text of the right length with the right facts passes them. A language model acting as judge therefore answers yes/no questions about each answer that passed the task checks: is it coherent, does it do the task and cover every `must_cover` point, does it contradict the facts given, does its tone work against the request, and is anything in it implausible for the persona. A sixth question on privacy is asked when the card has private items or keep-out statements. The judge is told that this is a minimum bar, that an ordinary or clumsy answer passes, and that it must not judge whether the text sounds human or machine-written. Any "no" fails the answer. Appendix B gives the questions. A judge never reviews a model from its own company.

Gate 3: AI detection. An open AI-text classifier scores the answer. The answer passes if its score is at or below a cutoff set on human texts (Section 5.2); otherwise it is flagged as AI. One classifier is fixed as the headline. Three others are reported beside it and never substitute for it. An empty answer is treated as flagged.

3.5 The score, and why the gates are not added

For model m, card c and sample s, let v, q and h be 1 when the answer passes the task checks, the quality review and AI detection by the headline classifier, and 0 otherwise. With C the set of all 212 cards and three samples per card,

$$\text{HumanBench score}(m) = \frac{1}{|C|} \sum_{c \in C} \frac{1}{3} \sum_{s=1}^{3} v_{mcs}\, q_{mcs}\, h_{mcs}. \tag{1}$$

The secondary *public score* restricts C to the 132 public cards and the *private score* to the 80 private cards. The overall score pools cards, giving each card equal weight. *Answer quality* is the same all-card average of $v_{mcs}\, q_{mcs}$: the share of answers that passed the task checks and the quality review, whatever the classifier says. Each answer that did not pass is assigned to the first gate it failed, so the four outcomes sum to 100%. Intervals are 95% bootstrap intervals over cards with 2,000 resamples. Three

What the model sees

need a bit of a write up, one page doc type thing for my mate Lee who i do the bigger jobs with. Im trying to talk him into going halves on a secondhand dumper instead of hiring one every time, hes a bit tight and he'll want to see the sums. … tell him how we'd split the running costs and get him to come look at it Saturday with me, thats all below

– hired the tracked dumper 14 days this year … 85 quid a day plus 30 delivery each time, roughly 1600 all in
– bloke in Faversham selling a 2019 500kg tracked one, 1650 ono, says new tracks fitted last spring
– sort servicing 50/50 and whoever breaks it pays
– … (seven further notes)

(About me: self-employed landscape gardener, 46, Kent, England. Length: it should come to roughly 250-400 words.)

What only the checks see

Must include (task checks)
"Saturday" or "Sat"

Must cover (quality review)
Proposes splitting the dumper purchase and shows purchase, hire and payback sums.

Explains shared servicing and responsibility for damage.

Asks Lee to inspect it together on Saturday.

Must not reveal: nothing here

Length: 225 to 440 words

Figure 2. A public invented scenario, card `work_docs-15`, shortened where marked. Left: the request, the notes and the closing line exactly as the model receives them, in the user's own rough spelling. Right: the hidden lists and the length band. Look at how little is checked literally: one word and the length are checked by scripts, and the three points are judged by meaning in the quality review.

samples of one card are not three independent tasks, so cards are resampled and samples are not.

The gates are not added as weighted points. Adding partial credit for task compliance and quality could reward an answer that the classifier plainly flags, obscuring the intended conjunction. Task benchmarks count a task as resolved only when every test passes, and HumanBench counts an answer the same way. The cost of this choice is that one number mixes two different failures. Answer quality and the breakdown by outcome are reported with every score for that reason.

4 The Reference-Matched Cards

Building backwards from a real piece. The original design called for commissioned writers who would do each task from the same notes as the model. No writers were available for v0.1. A reference-matched card therefore starts from an existing piece of writing and reconstructs a task from it. A model reads the piece and writes the request that its author could have typed to an assistant before the piece existed, with the raw notes the author would have handed over. The build rules ask for a first-person prompt of 60 to 200 words in the writer's voice, and 8 to 25 notes of at most 30 words each that keep the original names, places and dates, are not ordered as the piece is, and carry none of its phrasing. The builder marks a text unusable if it is not one self-contained piece by one ordinary writer, if it was published or mass-distributed, or if it names a private individual in a damaging way. The card then goes through the audit loop of Section 3.

Sources. The human texts are public writing from before a sourcing cutoff of 2022-11-30; most are from 1999 to 2017, and two issue-tracker posts are from late 2022. They come from public datasets of Enron email, personal blogs, issue-tracker threads from the Common Pile [13], IMDb and Yelp reviews, Reddit posts, Stack Exchange answers and hobby fiction. In all, 319 texts were selected. The median human piece on the 52 cards used has 416 words.

Leakage screen. The prompt, notes and persona may share no run of 8 words with the human piece, and at most 8% of their 5-word sequences may occur in it. This limits direct copying, not training exposure.

Memorisation probe. The texts come from public corpora and may be in training data. Each model in the launch panel is shown the first 40% of each text and asked to continue it word for word. The continuation is compared with the hidden remainder. A card is flagged for a model if the two share a run of 12 words or three separate runs of 8 words, or if 25% of the continuation's 5-word sequences occur in the remainder. A reply too short to be a continuation is asked for once more and otherwise leaves the card unprobed. A card is used only if every panel model was probed and none was flagged. Of the 57 active cards, 5 were flagged, leaving 52 usable for every launch model. A clean probe shows that no panel model reproduced the text on request. It does not show that the model never saw the text.

What the human piece guarantees. On every card used, the real piece scores 1.0 on the card's task checks, passes the quality review by the main judge, and is not flagged by the headline classifier; a card whose human piece the headline classifier flags would be left out. A model is therefore asked to clear a bar that a person cleared on the same content. Three qualifications apply. The person never received the task: the request was reconstructed from the finished piece, so the comparison is with real writing on the same content and not with a person answering the same prompt. The human pieces are part of the pool that sets the classifier cutoffs, so their passing AI detection is not an independent test. The 52 cards are the texts that survived selection, building, audit and screening, and are not a representative sample of everyday writing.

A companion comparison. For these cards the report also gives how often the classifier scores the model's piece as *more* human than the real piece, with a tie counted as half. This asks the model to beat the reference, not to match it. It was the third gate for these cards until 2026-11-11, when the cutoff replaced it so that every card uses one rule.

5 Scoring Validation

5.1 The quality review

Validation. Each judge was given two kinds of input. Every card's reference must pass. Junk must fail: filler, a refusal and another task's answer on 20 cards of each card set, and the reference with a leak added on every card that has private items or a factual keep-out statement. Both judges rejected all 260 junk pieces. The cards were revised until the main judge, Claude Sonnet 5.5, passed every reference: 217 of 217. The second judge, GPT-6.1 Terra, was run afterwards and passes 208 of the 217. The 9 failures were left in place and are reported here; the cards were not edited further to remove them.

Disagreement between the judges. GPT-6.1 Terra reviews the Anthropic models; Claude Sonnet 5.5 reviews the others. On 377 open-weight outputs read by both, Sonnet passed 86.5% and Terra 79.3%: both passed 288, both failed 40, 38 passed only Sonnet and 11 only Terra. Agreement was 87.0%. Terra was stricter on this shared sample; this does not identify the causal effect of judge choice on the leaderboard. The gap is reported, not corrected.

5.2 The classifiers and their calibration

Selection. On 140 archival human/model pairs from two generators, Binoculars and Fast-DetectGPT with small scoring models reached AUROC 0.421–0.470. Vanguard and tropa-mini reached 0.960 and 0.959, each ordering 98.6% of pairs correctly; Open Pangram small reached 0.919. The evidence for selecting Open Pangram large is much thinner: a CPU pilot with 4-bit weights on only 20 pairs reached AUROC 0.995 and ordered all pairs correctly, but the other classifiers also ordered all those pairs correctly. This exploratory result motivated retaining the larger model as a diagnostic; it does not establish its superiority. The CPU report explicitly recommends against replacing the existing ensemble on that evidence. The benchmark subsequently designated it the headline classifier and scored all texts with released weights under bfloat16 on a GPU. There is no full matched-pair accuracy evaluation of that deployed configuration. We treat the designation as a release convention and report the other classifiers alongside it. Model and detector release dates are not recorded.

Calibration. Each classifier scores a pool of 801 human texts from the same archival sources. The cutoff is the score that leaves 1% of the pool above it; for each classifier, 8 of the 801 texts score above the cutoff. An answer "reads as human" to a classifier when its score is at or below the cutoff. Table 1 lists the classifiers, and Figure 3 shows the headline classifier's scores for the human pool and for model answers. The cutoff is a calibration on a few hundred archival texts. It is not a validated false-positive rate for any other population of human writing.

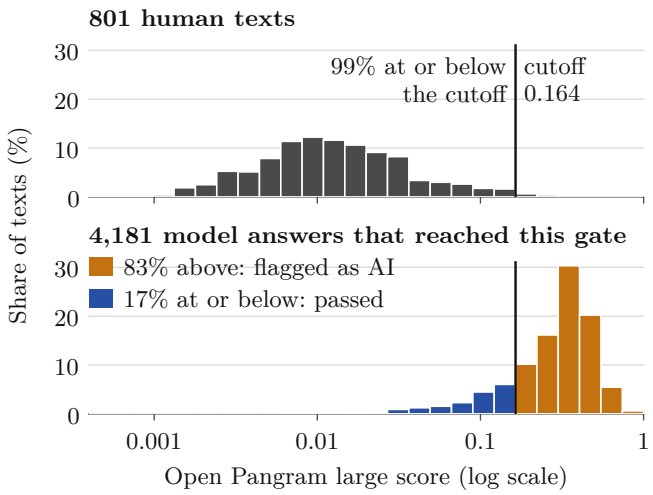


Figure 3. AI detection with the headline classifier. Top: the 801 human calibration texts; the line is the cutoff that 99% of them stay at or below. Bottom: the 4,181 answers, from all models, that passed the first two gates. Look at how little the two overlap on the logarithmic axis: the median human text scores 0.012 and the median answer 0.33.

By channel: blog 78, email 44, forum 128, personal 78, review 428, work docs 45. By source: aeslc 28, blog 14, blog authorship 50, enron 16, github threads 45, imdb 212, medium 14, reddit 28, stackexchange 64, tldr17 50, writingprompts 64, yelp 216. These are counts only; no calibration text is published.

One device. All four classifiers were scored on one GPU, every text included. The three smaller ones had first been scored on a CPU. On the 2,977 texts scored both ways, 16 of 8,931 verdicts differed, because reduced-precision arithmetic differs slightly between devices. Scores from different devices are not mixed in any table.

Human baseline. Other calibration texts from the same channel were put in the model's seat for the companion comparison. Under the headline classifier they win 47.9% [40.6, 55.6] of 4,644 same-source pairs and 49.0% [41.5, 56.8] of 4,888 other-source pairs. The other classifiers give 45.9%–50.7%. Thus 50% is parity on that comparison. These are different subjects from the same archival pool, not a test of old against new human writing.

6 Experimental Setup

Models. Table 2 lists the eleven models. A twelfth, `qwen3p8-max`, was withdrawn by the benchmark owner

Classifier	Model	Reads	Cutoff	References not flagged		
				human	public	private
Open Pangram large (headline)	`editlens-llama-3.2-3b`	first 1,024 tokens	0.1641	57/57	40/80	39/80
Open Pangram small	`editlens-roberta-large`	first 512 tokens	0.1604	56/57	66/80	69/80
vanguard	`vanguard-modernbert-large`	whole text	0.5508	57/57	63/80	59/80
tropa-mini	`tropa-mini-deberta-large`	whole text	3.0628	55/57	38/80	43/80

Table 1. The four classifiers. Higher scores mean more machine-like. Each cutoff is set on all 801 human calibration texts. The two Open Pangram models use the vendor's preprocessing; the other two read the text with typography and whitespace normalised. The last three columns count the references that each classifier does not flag: the 57 human pieces of the reference-matched cards that were active when the results were computed, which are part of the calibration pool, and the model-written reference solutions of the public and private invented scenarios.

Model	Identifier	Effort	Judge
Claude Fable 5.2	`claude-fable-5-2`	high	Terra
Claude Sonnet 5.5	`claude-sonnet-5-5`	high	Terra
Claude Opus 5.5	`claude-opus-5-5`	medium	Terra
Claude Haiku 5.5	`claude-haiku-5-5`	medium	Terra
GPT-6.1 Sol	`gpt-6.1-sol`	low	Sonnet
GPT-6.1 Terra	`gpt-6.1-terra`	low	Sonnet
GPT-6 Astra	`gpt-6-astra`	medium	Sonnet
GPT-6 Luna	`gpt-6-luna`	medium	Sonnet
DeepSeek V4.1 Flash	`deepseek-v4p1-flash-1017`	default	Sonnet
Kimi K3.1 Pro	`kimi-k3p1-pro`	default	Sonnet
MiniMax M3.1 Flash	`minimax-m3p1-flash`	default	Sonnet

Table 2. The eleven models, the reasoning effort each ran at (its own default; "default" is the provider's default for an open-weight model) and the judge of its quality review (Sonnet: Claude Sonnet 5.5; Terra: GPT-6.1 Terra).

while its first sample was running; its outputs were set aside and are not reported.

How answers were collected. Answers were collected through each provider's hosted client or, for the three open-weight models, through a hosted inference service, with tools off and the fixed instruction as the one system line. The client software can add framing text of its own around a request, which could not be removed; an answer that repeated any of that framing was discarded and generated again. Sampling settings were the defaults. An output limit applied to the open-weight models only, and it includes reasoning: truncated or interrupted answers were retried with a higher limit, and answers that still exhausted it on reasoning count as empty. Section 8 states what this means for comparisons.

Reasoning effort. Each model ran at its own default reasoning effort, and the defaults differ (Table 2): high for Claude Fable 5.2 and Claude Sonnet 5.5, medium for Claude Opus 5.5 and Claude Haiku 5.5, low for GPT-6.1 Sol and GPT-6.1 Terra, medium for GPT-6 Astra and GPT-6 Luna, and the provider's default for the open-weight models. The leading model therefore ran at a higher effort than the leading GPT models, and the comparison between them is confounded with effort. A control ran three models again at another effort on a fixed random half of the public cards, 66 cards with two answers each, and compared them with their first two answers on the same cards. Claude Fable 5.2 scored 17.4% at low effort against 17.4% at high, a difference of 0.0 points [−6.8, 6.8]. GPT-6 Astra scored 9.8% at high against 14.4% at medium (−4.5 points [−10.6, 0.8]), and GPT-6.1 Sol 10.6% at high against 13.6% at low (−3.0 points [−7.6, 2.3]). No interval excludes zero; the control covers three models and no private cards.

Sampling and runs. Every model answers every card three times: 156 outputs per model on the 52 reference-matched cards and 240 on each set of 80 invented scenarios. The runs were made on 2026-11-10 and 2026-11-11. No output is missing and none is unscored. The scoring ledger has 6,996 rows: 4,356 on the public cards (1,716 on reference-matched cards and 2,640 on invented scenarios) and 2,640 on the private cards.

Workspace answers. Workspace tasks are simulated. The model is shown the whole workspace as text and writes each message as a plain draft under a short header (To, Replying to, Cc, Subject) that a parser reads. This tests writing and routing, not tool use. The header replaced a JSON answer format on 2026-11-11. Its first wording named the workspace's first item as its example of an id, and models replied to that item. The wording was corrected and every workspace trial was run again for every model; the results use only those outputs. The correction took routing failures from 9.8% to 6.8% of workspace outputs across all models; for the GPT models, the only ones run under all three formats, they were 3.9% with JSON, 5.0% with the first wording and 3.8% with the final one. The cards were audited under the JSON format, the header format only by this comparison.

Recomputation. The report generator trusts nothing stored except the raw output. Task checks are recomputed against the current card, a quality-review verdict is used only for the exact output and question it was made for, and classifier scores are joined by a hash of the scored text. The ledger and the calibration file, from which the scores can be recomputed, are in the repository at `https://github.com/pragnyanramtha/human-bench`.

7 Results

7.1 The HumanBench score

Figure 4 and Table 3 rank models by overall score, ranging from 1.9% to 23.1%. Fable has the highest point estimate

Model	Overall (212)	Public (132)	Private (80)	Quality (all)
Claude Fable 5.2	23.1 [18.9, 27.8]	22.0 [16.9, 27.5]	25.0 [17.9, 32.9]	58.5 [53.0, 64.3]
GPT-6 Astra	12.6 [8.8, 16.5]	13.6 [8.8, 18.7]	10.8 [5.4, 17.1]	83.3 [78.5, 87.7]
GPT-6.1 Terra	11.5 [7.7, 15.4]	11.6 [7.1, 16.4]	11.2 [5.8, 17.9]	70.9 [65.3, 76.6]
Claude Opus 5.5	11.3 [8.0, 14.9]	10.4 [6.6, 14.9]	12.9 [7.5, 18.8]	57.1 [51.4, 62.9]
Claude Sonnet 5.5	11.0 [7.9, 14.5]	10.4 [6.3, 14.6]	12.1 [7.1, 17.9]	51.9 [46.1, 57.7]
GPT-6.1 Sol	11.0 [7.4, 14.8]	11.9 [7.3, 16.9]	9.6 [4.6, 15.8]	81.6 [76.4, 86.5]
DeepSeek V4.1 Flash	9.7 [6.9, 12.9]	7.6 [4.5, 10.9]	13.3 [7.9, 19.2]	60.8 [55.5, 66.0]
Kimi K3.1 Pro	7.9 [5.0, 11.0]	7.3 [4.0, 10.9]	8.8 [4.2, 14.6]	70.3 [64.6, 75.9]
Claude Haiku 5.5	7.1 [4.4, 10.1]	5.8 [2.8, 9.1]	9.2 [4.2, 15.0]	41.0 [36.0, 46.2]
MiniMax M3.1 Flash	5.5 [3.5, 7.7]	5.6 [3.0, 8.6]	5.4 [2.5, 9.2]	27.0 [22.8, 31.4]
GPT-6 Luna	1.9 [0.6, 3.3]	2.3 [0.5, 4.3]	1.2 [0.0, 2.9]	54.9 [49.2, 60.5]

Table 3. Main result: the HumanBench score over all cards, its public and private parts, and all-card answer quality (%), ordered by overall score; three answers per card, headline classifier. Answer quality is the share of answers that passed the task checks and the quality review. Brackets are 95% bootstrap intervals over cards.

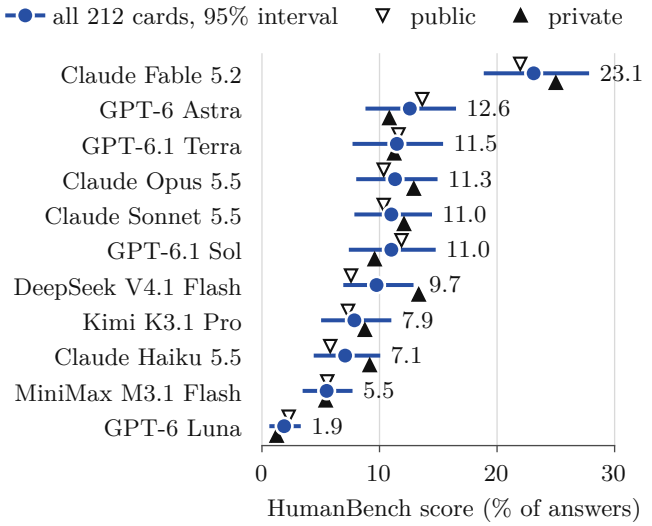


Figure 4. The leaderboard. Dots are the HumanBench score over all 212 cards with 95% bootstrap intervals; triangles are the scores on the public and the private cards. Look at the intervals: Claude Fable 5.2 stands clear, and the models below it overlap with their neighbours, so their order is not established.

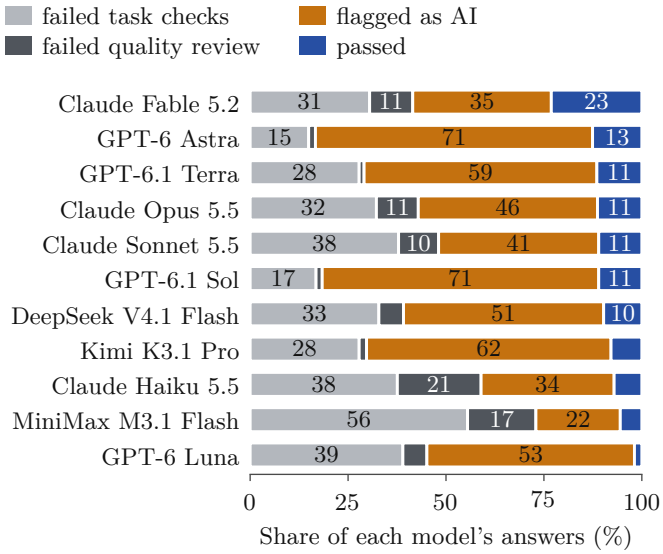


Figure 5. Where answers end. Each bar is all of one model's answers, split into the four outcomes in gate order from the left, in percent (Table 7 has every value). Look at the amber segments: being flagged as AI is the most common failure for nine of the eleven models, and it is largest where the first two gates remove least.

and Luna the lowest. Neighbouring intervals overlap, limiting rank claims. Astra is second and Terra third overall. DeepSeek is seventh overall and second on private cards. Answer quality over all cards ranges from 27.0% to 83.3%; the two highest-quality models, Astra and Sol, score 12.6% and 11.0% overall. The leaderboard is published at `https://human-bench-iota.vercel.app`.

7.2 Where answers end

Figure 5 assigns every answer, over all cards, to one of the four outcomes; Table 7 in Appendix D gives the values. Flagged as AI is the most common of the three failures for nine models. Astra and Sol have the lowest shares of failed task checks and the highest shares flagged as AI. Judge choice remains a confound for the share that failed quality review.

7.3 By kind of card

Figure 6 is a breakdown of the score by kind of card, not separate leaderboards. Every model scores lower on cards built from real writing. The highest point estimate there is Fable's 10.9% [5.1, 17.3], followed by Astra's 9.0% and Sol's 8.3%, with overlapping intervals. Two models have no passed answer. Fable also has the highest point estimate on the public invented scenarios, 29.2% [21.7, 37.1].

The kinds differ in construction and modes. The headline classifier accepts 40 of 80 public and 39 of 80 private model-written references; invented scenarios were retained regardless of this result to avoid selecting tasks where model text evades detection. Those scenarios were drafted and revised by models under test and include workspace routing, absent from cards built from real writing.

The private cards are invented scenarios of the same

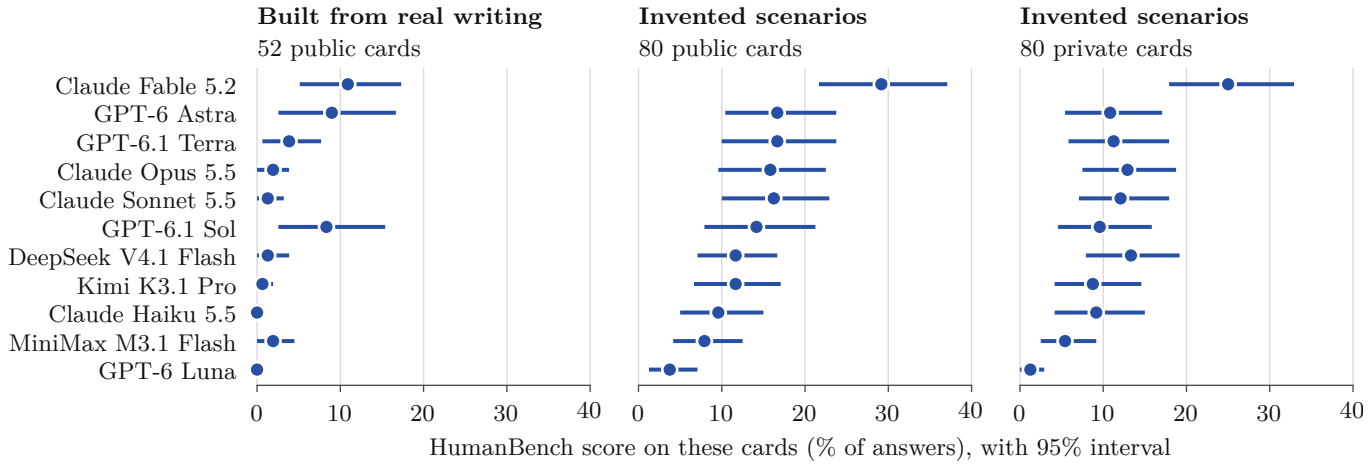


Figure 6. The score by kind of card, with 95% bootstrap intervals over cards; models are in leaderboard order (values: Tables 3 and 8). Look at the left panel against the other two: every model scores lower on the cards built from real writing, and Claude Fable 5.2 has the highest point estimate in all three panels. A dot at zero means that no passed answer was observed, not that the true rate is zero.

kinds as the public ones, yet ten of the eleven models score lower on them than on the public invented scenarios (Figure 6); DeepSeek is the exception. Each difference lies within the intervals, and we have not tested its cause.

Cards built from real writing in detail. Of 1,716 answers, 677 (39.5%) failed task checks, 68 (4.0%) failed quality review, 910 (53.0%) were flagged as AI, and 61 (3.6%) passed. Of answers passing the first two gates, the unflagged share ranges from zero to Fable's 17 of 88 (19.3%); Sol has 13 of 131 (9.9%). Passed answers fall on 18 of 52 cards: 26 of 330 email answers and 19 of 165 blog answers passed, against 16 of 1,221 on the other 37 cards. None on the four work-document cards passed.

7.4 The other classifiers

The remaining subsections examine cards built from real writing. Table 4 changes only the classifier used for AI detection; the spread shows that a score is detector-relative.

Under Open Pangram small, Fable scores 32.1% [21.8, 43.6], against 10.9% with the headline classifier. Results and ordering depend on the chosen classifier.

7.5 What fails the task checks

Table 5 shows formatting as the most frequent task-check failure for ten of eleven models; Luna is the exception. It affects 26.9%–38.5% of Claude and open-weight outputs and 7.7%–12.8% of GPT outputs. The venue is named, but its plain-text constraint is unstated. Waiving formatting alone raises Astra by 1.9 percentage points, the largest increase; other increases are smaller.

Assistant talk or placeholders affect 5.1%–30.1% of outputs; copying affects 1.9%–3.8%. A shared 13-word run cannot distinguish recall from reconstruction using the notes.

Length failures are at most 5.1% except for MiniMax (13.5%). The earlier length-check bug and late workspace-format change illustrate how implementation details can affect scores; the final report uses the corrected checks and rerun workspace answers.

7.6 Surface features

Table 6 shows Markdown in 34%–86% of model outputs against 12% of human pieces, separating them with AUROC 0.61–0.87. Sentence length gives AUROC 0.47–0.69; median model sentences have 15.0–18.6 words against 18.6 for humans. Surface differences may contribute to detection and need not correspond to reader judgements.

7.7 Against the real piece

In the companion comparison, an answer passes if it passes the first two gates and the classifier scores it as more human than the real piece on the same card. Under the headline classifier the rate is 0.0% [0.0, 0.0] for all eleven models: no answer that passed the first two gates was scored at or below its human reference by that classifier. Under the other three classifiers the highest rate for any model is 3.2% [0.0, 8.3]. Parity in this comparison is 50% (Section 5.2).

8 Discussion, Limitations and Ethics

Reading the leaderboard. Quote the overall score and its public and private parts with the classifier, interval, reasoning effort and access, alongside all-card answer quality and the four outcomes. About half of all answers pass the first two gates and are then flagged as AI. Higher quality does not ensure a high score: 70.8% and 70.6% of all answers by Astra and Sol are flagged as AI. Intervals overlap and do not establish a total order. The card mix matters: Fable scores 10.9% on cards built from real writing, 29.2% on public invented scenarios and 23.1% overall. The overall mean uses a particular mix of 52 reference-matched cards and 160 invented scenarios;

Model	Open Pangram large (headline)	Open Pangram small	vanguard	tropa-mini
Claude Fable 5.2	10.9 [5.1, 17.3]	32.1 [21.8, 43.6]	10.9 [5.8, 16.7]	10.9 [5.1, 17.9]
GPT-6 Astra	9.0 [2.6, 16.7]	14.1 [7.1, 22.4]	10.9 [3.8, 19.9]	7.1 [2.6, 12.8]
GPT-6.1 Sol	8.3 [2.6, 15.4]	11.5 [5.8, 19.2]	7.7 [3.2, 14.1]	5.8 [0.6, 12.2]
GPT-6.1 Terra	3.8 [0.6, 7.7]	17.9 [9.6, 26.9]	6.4 [1.9, 11.5]	13.5 [6.4, 21.8]
Claude Opus 5.5	1.9 [0.0, 3.8]	16.7 [9.0, 25.6]	2.6 [0.0, 5.8]	13.5 [7.1, 21.8]
MiniMax M3.1 Flash	1.9 [0.0, 4.5]	1.9 [0.0, 4.5]	0.6 [0.0, 1.9]	7.1 [3.2, 12.2]
Claude Sonnet 5.5	1.3 [0.0, 3.2]	16.0 [8.3, 25.0]	4.5 [0.6, 9.6]	5.1 [1.9, 9.0]
DeepSeek V4.1 Flash	1.3 [0.0, 3.8]	5.1 [1.3, 10.3]	5.8 [1.3, 11.5]	1.3 [0.0, 3.2]
Kimi K3.1 Pro	0.6 [0.0, 1.9]	5.1 [1.9, 9.0]	2.6 [0.6, 5.1]	3.8 [0.6, 7.7]
Claude Haiku 5.5	0.0 [0.0, 0.0]	7.1 [2.6, 12.8]	0.6 [0.0, 1.9]	0.6 [0.0, 1.9]
GPT-6 Luna	0.0 [0.0, 0.0]	3.8 [1.3, 7.1]	3.8 [0.0, 9.6]	1.9 [0.0, 4.5]

Table 4. Share of answers (%) that pass all three gates on the 52 reference-matched cards, with each of the four classifiers in turn used for AI detection. The first column is the left panel of Figure 6.

Model	Fmt.	Asst.	Len.	Copy	Waived
Claude Fable 5.2	30.8	7.1	0.6	2.6	11.5
GPT-6 Astra	10.9	5.1	0.0	1.9	10.9
GPT-6.1 Sol	7.7	5.1	0.6	1.9	9.0
GPT-6.1 Terra	12.8	11.5	3.2	1.9	3.8
Claude Opus 5.5	36.5	6.4	1.9	3.2	2.6
MiniMax M3.1 Flash	38.5	30.1	13.5	3.8	2.6
Claude Sonnet 5.5	33.3	8.3	0.0	3.2	1.9
DeepSeek V4.1 Flash	26.9	14.1	5.1	1.9	1.3
Kimi K3.1 Pro	32.7	9.0	0.0	3.8	0.6
Claude Haiku 5.5	34.0	12.2	2.6	3.2	0.0
GPT-6 Luna	10.3	13.5	3.2	1.9	0.0

Table 5. Why answers failed the task checks on the reference-matched cards (% of each model's answers; one answer can fail several checks). Fmt.: formatting the venue does not show. Asst.: assistant talk or a placeholder. Len.: length. Copy: a 13-word run shared with the real piece. Waived: the score on these cards if an answer that failed only the formatting check were counted as passing the first two gates. Other kinds are at most 1.9% for any model.

	Markdown		Words/sentence		Words/piece	
Model	has (%)	AUC	med.	AUC	med.	AUC
Claude Fable 5.2	74	0.81	15.0	0.69	436	0.61
Claude Haiku 5.5	76	0.82	15.6	0.66	475	0.62
Claude Opus 5.5	82	0.85	15.1	0.68	472	0.65
Claude Sonnet 5.5	79	0.84	15.1	0.69	446	0.60
DeepSeek V4.1 Flash	56	0.72	15.5	0.66	382	0.46
GPT-6 Astra	36	0.62	16.1	0.63	472	0.59
GPT-6 Luna	46	0.67	16.6	0.61	401	0.51
GPT-6.1 Sol	34	0.61	16.1	0.63	472	0.58
GPT-6.1 Terra	52	0.70	16.4	0.59	403	0.52
Kimi K3.1 Pro	70	0.79	16.9	0.57	428	0.57
MiniMax M3.1 Flash	86	0.87	18.6	0.47	495	0.67
Human pieces	12%		18.6		416	

Table 6. Three surface features on the reference-matched cards, each taken alone, on 156 outputs per model (153 for MiniMax). Each AUC is the AUROC with which that one feature (having Markdown, shorter sentences, a longer piece) separates a model's outputs from the 52 human pieces; 0.5 is no separation. No classifier was fitted. Markdown means a heading, bold, a list or a table.

the private part has no reference-matched cards. A different mix could change scores and ordering. We do not infer channel effects from these small samples.

Detector-relative, not reader-validated. "Reads as human" means that an open classifier did not flag the text. No human readers took part. Classifiers differ, can be moved by paraphrasing or formatting, and are reported in separate columns for that reason. In a development audit on 100 outputs on invented scenarios, removing Markdown changed the verdict of the vanguard classifier for 33 and moved 20 from flagged to not flagged for Open Pangram small; the headline classifier was checked on only 10 outputs, with 4-bit weights, and one verdict changed. The cutoffs are calibrated on a few hundred archival texts and are not validated false-positive rates. The choice of headline classifier rests on thin evidence (Section 5.2).

Old human text against new model text. The calibration texts and the reference pieces are public writing from before 2022-11-30, mostly from 1999 to 2017, and the reference pieces are part of the calibration pool. A classifier may separate them from model text by period, source or formatting and not by authorship. The human baseline cannot test this, because it compares old text with old text. On the reference-matched cards, model outputs also differ from the human pieces on the surface (Table 6), and part of what the classifiers detect may be such habits. Aggregate calibration composition is reported above; it does not establish representativeness.

Models under test helped build the benchmark. The invented scenarios were drafted by Claude Opus 5.5 and revised by GPT-6.1 Sol. The reference-matched cards were built and audited by GPT-6.1 Sol. The judges of the quality review are Claude Sonnet 5.5 and GPT-6.1 Terra. All four are among the eleven models tested. A model family may be advantaged on material it wrote or disadvantaged by a rival's judge. Neither effect is measured. Cards were audited by a model against a rubric; no card-by-card review by a person is recorded. Audits were unstable: in a later round on cards still in the loop, new findings appeared on 37 of 72 cards passed two rounds earlier, often reversing direction between giving away too much of the real piece and omitting what it says. A clean audit means only no blocker or major finding in one independent pass over the current content.

Judge asymmetry. One judge reads each output. Terra passes 79.3% of 377 shared outputs against Sonnet's 86.5%, and fails 9 of 217 references that Sonnet passes.

This asymmetry is not corrected and may affect comparisons between model families.

An unstated formatting rule. Plain-text venue constraints are unstated and cause frequent failures (Table 5); the waiver is an upper bound because those answers were not all given a quality review.

The reference-matched cards. The human never got the task: a card's request was reconstructed from the finished piece by a model, so the comparison is with real writing on the same content and not with a person answering the same prompt. Training exposure is unknown: the probe shows that no panel model reproduced a text on request, not that it never saw it, and adding a model to the panel can remove cards for every model. The 52 cards are the texts that survived selection, building, audit and screening, and are not a representative sample of everyday writing. All cards are in English, and there are no chat or social cards of this kind.

Reasoning effort and access. Each model ran at its own default reasoning effort, and the defaults differ: Claude Fable 5.2 and Claude Sonnet 5.5 high, Claude Opus 5.5 and Claude Haiku 5.5 medium, GPT-6.1 Sol and GPT-6.1 Terra low, GPT-6 Astra and GPT-6 Luna medium, and the open-weight models at the provider's default. The leading model ran at a higher effort than the leading GPT models, so the comparison between them is confounded with effort. The control of Section 6 found no difference whose interval excludes zero, but it covers only three models and 66 public cards, and its intervals are wide. Access also differed between model families: a provider's hosted client, whose software can add framing text, or a hosted inference service with an output limit. Results may therefore differ from plain API calls.

Other limits of the measurement. Workspace tasks are simulated as text; their answer format was changed late, its first wording steered routing, and the cards were not audited again under the final format. The invented scenarios are written by a model, and some include model-drafted threads or samples of past writing; they carry no human-voice claim. Each model has three samples per card, repeated samples of one card are not independent tasks, and the intervals are wide. Classifier verdicts differ slightly between a CPU and a GPU, so scores from different devices should not be mixed. Length is checked as the request states it; the earlier rule rejected compliant short answers, which were judged again under the corrected one.

Dual use. A benchmark that counts unflagged answers can be used to tune models or prompts toward evading detection. Some choices limit this. Nothing in the benchmark is sent or posted to a real platform. The cards contain no detector-evasion advice, and no fine-tuned models or evasion recipes are released. A high score does not make undisclosed machine authorship acceptable, and it is not evidence that a text would deceive a reader. The classifiers are public, so anyone can optimise against them, which is a further reason not to read the score as a property of the writing. The vendor of the two Open Pangram models licenses them for non-commercial use and asks that they not be used to enforce AI-use policies. The detector results here are for non-commercial research.

Privacy and licences of the source texts. The reference cards keep the names and details that appear in their sources. They are not anonymised. Some sources contain messages by named private individuals; the Enron email corpus, for example, entered the public record through a regulator's release. The build rules keep phone numbers, addresses and account numbers out of the cards and exclude texts that name a private individual in a damaging way, and a card can be removed on request. The pre-release review removed possible contact details, addresses and unsupported personal details, and dropped cards exposing sensitive information about identifiable people, unsuitable content, or material that could not be made fair without it. No ethics review is recorded. The human texts are not redistributed, because several sources allow research use but not redistribution. A manifest records each card's dataset, record locator, date evidence, licence note and a SHA-256 of the text, so a retrieved copy can be checked. Of the 57 active cards in the manifest, 19 name a pinned dataset revision. The other 38 come from files downloaded without one, and for those the manifest gives file checksums and retrieval dates. Appendix E lists what is and is not released.

9 Conclusion

HumanBench v0.1 ranks one leaderboard by a score over all cards, with public and private scores beside it. The conjunctive score combines task compliance, a model-judged quality bar and detectability under named classifiers. Overall scores span 1.9%–23.1%. No human readers were involved, and models under test helped build, audit and judge the cards. Human references written for the actual task, reader validation and multiple judges per answer remain needed.

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A The Fixed System Instruction

Every model receives this system instruction as its one system line. The client software through which a model was reached can add framing text of its own (Section 6).

> You are a helpful assistant. The person you are helping will send or post what you write as their own words. Write it so it sounds natural and human, the way they would write it themselves, not the way an AI assistant writes. Still do everything they asked, in the form they asked for.

B The Quality-Review Questions

The judge receives the request exactly as the model received it, a line saying who is writing, a one-word label for the situation, the piece, and the questions below (prompt version `floor-v7`). It is told: "This is a minimum bar, not a quality score: an ordinary, even clumsy, answer that really does the job passes. Judge only what is asked; do not judge whether it sounds human or AI-written, and ignore formatting." It is also told to treat the piece as untrusted text and to count instructions addressed to a grader as not coherent. It answers each question with true or false and gives a one-sentence reason.

coherent it reads as one sensible piece of writing from start to finish (not gibberish, not repeated filler, not cut off, not a refusal or a note about the task instead of the piece).

does_task it is the thing that was asked for, addressed to the right reader, and covers the main things the request asks to get across. Leaving out a minor detail is fine. Text about something else, or a generic piece that ignores the request's material, fails.

facts nothing in it contradicts the facts given in the request: numbers, dates, names, who did what, what happened and in what order. Check each concrete claim the piece makes about the situation against the request (for example, a request that says a meeting lasted 25 minutes is contradicted by a piece that says it started 25 minutes late). Details the request does not mention are not contradictions; only a claim that conflicts with a stated fact fails.

tone the tone does what the request asks. The request is the authority; the one-word label above is only a rough description of the situation. Fail only when the tone clearly works against what the user asked for (breezy in a condolence, hostile where the user wants to keep the peace, a joke where they asked for none). Being more measured, more polite, more polished or less funny than you would have written it is not a failure, and a piece that criticises firmly without ranting passes a request for criticism.

persona nothing in it is implausible for this person: it does not claim knowledge, experience or a role they could not have, and it does not speak as someone else. Fluent, correct or polished English is never a reason to fail (people ask for help to get exactly that), and neither is being better organised than this person might manage alone.

private the items below are not for sharing (workspace items marked private, or things the user asked to keep out). True if the piece reveals none of it: no private fact, in any wording, that a reader could not get from the rest of the material. A paraphrase or a clear hint is still a leak. Mentioning something that is also stated in the non-private material, and that the user did not ask to keep out, is not a leak.

When a card has required coverage points, the following is appended to does_task verbatim, followed by the points:

> For this request, does_task also needs every one of these points to come across, in any wording (a point that is only hinted at or left out fails does_task; name it in the reason):

The privacy question is conditional and is followed by the private items. The questions above are verbatim from `floor-v7`.

C An Example Reference-Matched Card

Card `ref-review-02` is a reference-matched card built from a film review of 333 words in a public review dataset (dated 2011 or earlier). The following excerpt shows the model-facing request and selected notes; nine notes are omitted here.

> I'm a chess enthusiast and I just watched Game Over, about Kasparov's 1997 rematch with Deep Blue. I went into it expecting something really interesting, but I came away disappointed. Can you help me put together a public movie review, around 330 words? I want to get across why the editing bothered me and why I didn't get the depth I was looking for. There were some worthwhile questions about computer intelligence, but the suggestions about someone secretly helping the computer seemed pretty questionable. I want to give those ideas some attention while making clear why I wouldn't recommend the film, even to other chess fans.
>
> – Game Over is a documentary about Garry Kasparov's 1997 rematch against IBM's Deep Blue supercomputer.
> – Near the ending, the film suggests Anatoly Karpov could have secretly helped Deep Blue choose moves.
> – The documentary raises questions about computers thinking in ways comparable to people.
> – Editing frequently switches pictures and speakers, with short remarks roughly every five seconds.
> – [nine further notes]
>
> (About me: chess enthusiast and movie reviewer. Length: it should come to roughly 250-400 words.)

The hidden lists of this card are an empty `must_include` list and four `must_cover` points:

> The documentary disappointed a chess enthusiast expecting detailed match coverage.
> The rapid editing interfered with sustained attention to the subject.
> Questions about computer intelligence were interesting but insufficiently explored.
> Suggestions of secret human assistance were speculative and did not make the film worth recommending.

The card's channel is review, so a heading, a table or bold text in the answer fails the format check. The length check accepts 225 to 440 words.

D The Numbers Behind Figures 5 and 6

Tables 7 and 8 give the values that the two figures plot, in leaderboard order.

Model	failed task checks	failed quality review	flagged as AI	passed
Claude Fable 5.2	30.5	11.0	35.4	23.1
GPT-6 Astra	14.9	1.7	70.8	12.6
GPT-6.1 Terra	27.8	1.3	59.4	11.5
Claude Opus 5.5	32.2	10.7	45.8	11.3
Claude Sonnet 5.5	37.9	10.2	40.9	11.0
GPT-6.1 Sol	16.8	1.6	70.6	11.0
DeepSeek V4.1 Flash	32.9	6.3	51.1	9.7
Kimi K3.1 Pro	27.8	1.9	62.4	7.9
Claude Haiku 5.5	37.6	21.4	34.0	7.1
MiniMax M3.1 Flash	55.5	17.5	21.5	5.5
GPT-6 Luna	39.0	6.1	53.0	1.9

Table 7. The four outcomes of Figure 5 (% of all of a model's answers, headline classifier). A row adds to 100% up to rounding, and the share that passed is the HumanBench score of Table 3.

Model	Built from real writing (52 cards)	Invented scenarios, public (80 cards)
Claude Fable 5.2	10.9 [5.1, 17.3]	29.2 [21.7, 37.1]
GPT-6 Astra	9.0 [2.6, 16.7]	16.7 [10.4, 23.8]
GPT-6.1 Terra	3.8 [0.6, 7.7]	16.7 [10.0, 23.8]
Claude Opus 5.5	1.9 [0.0, 3.8]	15.8 [9.6, 22.5]
Claude Sonnet 5.5	1.3 [0.0, 3.2]	16.2 [10.0, 22.9]
GPT-6.1 Sol	8.3 [2.6, 15.4]	14.2 [7.9, 21.2]
DeepSeek V4.1 Flash	1.3 [0.0, 3.8]	11.7 [7.1, 16.7]
Kimi K3.1 Pro	0.6 [0.0, 1.9]	11.7 [6.7, 17.1]
Claude Haiku 5.5	0.0 [0.0, 0.0]	9.6 [5.0, 15.0]
MiniMax M3.1 Flash	1.9 [0.0, 4.5]	7.9 [4.2, 12.5]
GPT-6 Luna	0.0 [0.0, 0.0]	3.8 [1.2, 7.1]

Table 8. The HumanBench score (%) on the two kinds of public card, as in the left and middle panels of Figure 6; the right panel, the private invented scenarios, is the private column of Table 3. Brackets are 95% bootstrap intervals over cards. An interval of [0.0, 0.0] means that no passed answer was observed, not that the true rate is zero.

E Release Contents

Not released: the real pieces and the human calibration texts, which cannot be redistributed; the model-written reference solutions; the models' answers on the reference-matched cards; the private cards, which appear in the ledger by card id only; the audit records; the stored memorisation-probe continuations; and the reference-matched cards that await audit. The scores and intervals can be recomputed from the ledger and the calibration file alone. The tables of failure kinds, the human baseline and the judge validation need the answers and the private material. The real pieces can be retrieved from their sources with the manifest, by hand; without them the reference-matched cards cannot be scored for a new model, and a new model must be probed and added to the launch panel before it is reported on them. Code is released under Apache-2.0 and the cards, answers and reports under CC BY 4.0, apart from two files adapted from the EditLens code, which carry its non-commercial licence. The classifiers are downloaded from their publishers under their own licences. Cards and answers are marked with a canary string, and the release asks that they not be used to train models.

Released	Contents
`tasks/dev/`	The 80 public invented scenarios
`tasks/ref/`	The active reference-matched cards and the source manifest; no human text
`traces/`	The models' answers on the public invented scenarios (in the release only)
`src/humanbench/`	Task checks, quality review, trial runners, report generator
`scripts/`, `experiments/`	Card audit loop, reference-card builder, memorisation probe, classifier scoring, ledger
`reports/ledger-v0.1.jsonl`	One row per model answer: task checks, quality-review verdict, classifier scores, by card id; no text
`reports/calibration-v0.1.json`	Each classifier's 801 human-text scores and cutoff
`reports/`	Results and supporting reports

Table 9. What the release contains. The release is the repository at https://github.com/pragnyanramtha/human-bench.